

SoNDe CB

Application Note

Compatible with
SoM firmware v0r11
FPGA firmware v0r10

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1 Applicable Documents

[IDE-TB-M-BV0.26]	IDEAS Testbench Software Reference Manual
[IDE-REP-Ref-V1.7]	IDEAS Readout and Control Packet Protocol Reference

2 Revision History

Date	Version	Author	Comment
2019-03-22	VOR0	Arne Fredriksen	Initial version
2019-04-01	VOR1	Arne Fredriksen	Added info about the data formats
2019-05-08	VOR2	Arne Fredriksen	Added temperature sensor, one-channel readout
2019-05-29	VOR3	Arne Fredriksen	-Fixed bug in ASIC ID column in the channel to pixel conversion table -Expanded on the introduction -Fixed channel ID field description in trigger time mode
2019-05-31	VOR4	Arne Fredriksen	-Added info about the new dummy data mode
2019-11-26	VOR5	Arne Fredriksen	-Added info about timestamp wrap, sync, SoC temperature sensor
2021-03-18	VOR6	Arne Fredriksen	-Added info about Sync/MLVDS -Updated system register table
2022-10-28	VOR7	Ranjisha R	-Updated channel id details for wrap-around and timestamp reset data packet
2024-06-03	VOR8	Codin Gheorghe	System register settings updated
2024-06-26	VOR9	Codin Gheorghe	Dummy Data Generator updated

3 Introduction

The SoNDe CB (Solid-State Neutron Detector Concentrator Board) connects to four S-DAM front-end boards, each containing four 16-channel readout ICs from IDEAS. In total, 256 channels are read out with Hamamatsu H12700/H8500 MaPMTs. The board contains a processor board containing a system-on-chip, in addition to an FPGA. A photograph of the SoNDe CB with four front-end boards attached is shown in Figure 1.

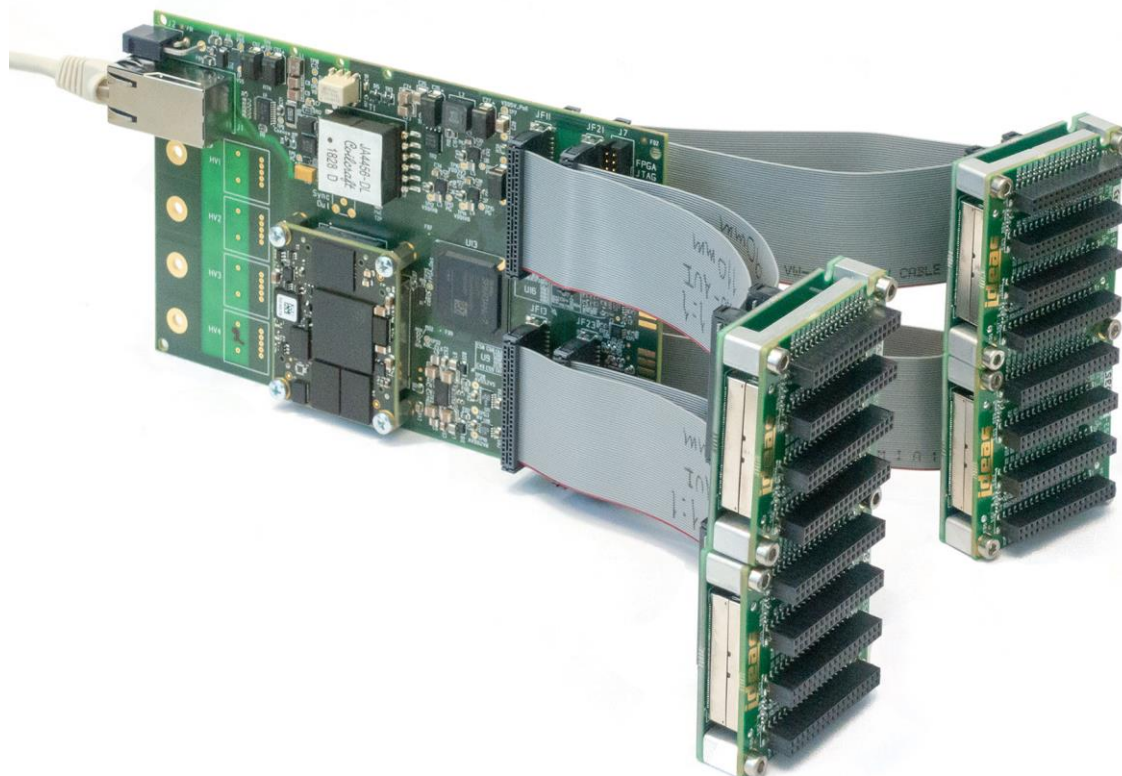


Figure 1: SoNDe CB with front-ends.

Figure 2 shows an overview of the concentrator board and its relation to connected boards. Below is a list of the main parts with brief descriptions.

- **Concentrator board**
 - **System on Module**, used for communication and data acquisition.
 - **FPGA**, used for handling triggers from the front-ends.
 - **High voltage**, up to -1200V for each front-end.
 - **Calibration pulse generator**, delivering test charges to all front-ends.
 - **ADC**, 14-bit pipeline ADC, reading from one front-end at a time.
 - **Sync in/out**, SMB connectors for syncing between boards.
 - **Sync MLVDS**, Micro-Match connector for syncing between boards.
 - **Power**, either Power-over-Ethernet through RJ45, or 22-57 VDC through Mini-Fit JR.
- **Front-end board**

- **IDE3465**, 16-channel ASIC with charge sensitive preamplifiers and individual trigger outputs for each channel. 4 daisy-chained ASICs per front-end.
- **Temperature sensor**, TMP112 sensor, located beneath the ASIC cap.
- **Interface board**, accommodates front-end board signals for the concentrator board.

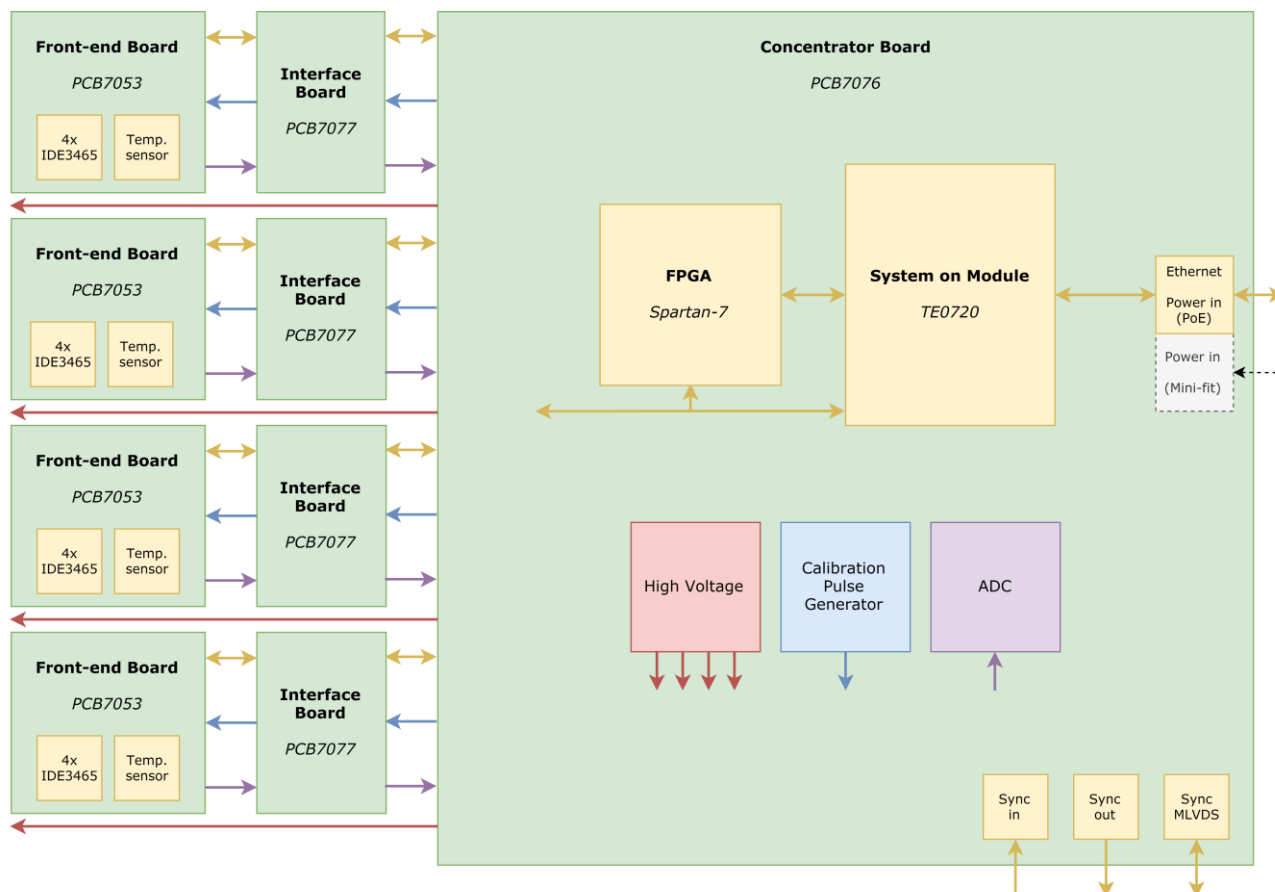


Figure 2: Block diagram of SoNDe CB with front-ends.

4 Operation

4.1 Operational Modes

The SoNDe CB can operate in three modes:

- In **trigger time** mode, you receive the timestamp and channel number of the triggered channels. Several channels can trigger simultaneously.
- In **all-channel readout**, all 256 channels are read out spectroscopically at the same time. This readout can either be activated by charges detected on the inputs, or by the on-board calibration pulse generator.
- In **one-channel readout**, one channel is read out spectroscopically upon detecting a charge over threshold for that channel.

4.2 Readout Control

The back-end registers are listed in Table 7 in section 5.2.

Four registers are used to control the readout, assuming that the front-ends have been configured properly:

Table 1: Readout control registers.

Register name	Description
Readout enable	3-bit register enables either the trigger time mode or the pulse-height mode: • 0b011 enables trigger time • 0b101 enables all-channel readout • 0b111 enables one-channel readout • 0b000 disables both modes
Readout reset	1-bit register that resets the readout. Internal memories in the FPGAs are cleared, and state machines are reset. Active high.
Reset timestamp	1-bit register that forces the timestamp counter back to zero. When issued, a timestamp reset data packet is transmitted, as described above.
Dummy data generator	2-bit register that enables dummy data transmission: • 0b01 enables fixed channel with increased ADC values • 0b10 enables ramped channel with fixed ADC value • 0b00 disables the dummy data Must be used with one of the readout modes enabled.
Hold delay	16-bit register that sets the hold delay in all-channel readout. This value must be set differently depending on whether the inputs are read out, or if the calibration charge injector is used. A hold delay of 1.45 μ s is recommended. This is achieved using the register value 292 (calibration), 264 (regular readout). Increments 5 ns per LSB.
Trigger time number of events	16-bit register that sets the number of events that are collated in a trigger time data packet, for increased data rate capability. Maximum 250.

4.3 Calibration Pulse Generator

The on-board calibration pulse generator is controlled with the registers in the 'Cal Pulse Gen' category in Table 7. The pulse interval must be longer than the pulse length. If the number of pulses is set to 0, an infinite set of pulses is sent, which can be cancelled by setting the number to another value, like for instance 1. Pulses with the configured settings are sent when a '1' is entered into the 'Calibration execute' register. Cal pulsing works with all-channel-readout and trigger time readout.

The amplitude of the pulses is set by the 'Calibration Pulse DAC' register.

4.4 HV Bias

There is a HV bias for each front-end, controlled with the 12-bit HV DACs through the system registers. The range is from 0 V to -1200 V (without load). Figure 3 shows the DAC-to-voltage-relationship, based on measurements. One LSB corresponds to approximately 295 mV. Small deviations from this will apply.

There is no safeguard against changing the bias too fast. Please use caution when adjusting the HV bias DAC values.

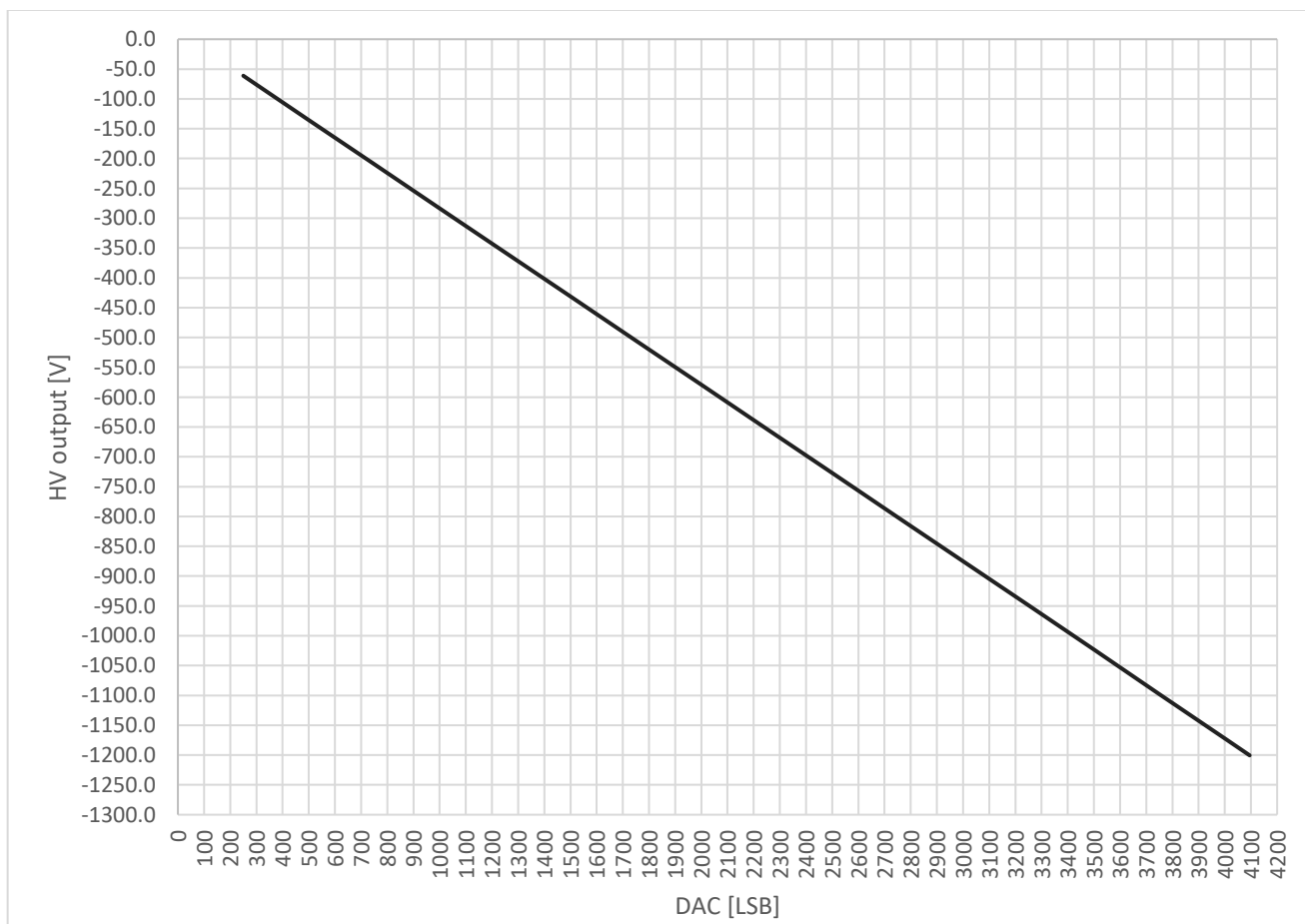


Figure 3: Typical DAC value versus HV output.

4.5 Temperature Monitor

Each front-end has a TMP112A temperature sensor, located close to the ASICs beneath the ASIC lid. These can be read out by the system registers in the *Monitoring* category, where there is one register for each front-end temperature sensor.

When a read operation on a temperature sensor's system register is performed, the raw data from the sensor is transmitted. By shifting this number 4 bits to the right and then multiplying by 0.0625, you get the (uncalibrated) temperature in degrees Celsius.

There is also a temperature monitor on the SoC (System on Chip) that is located on the SoNDe CB. This temperature sensor is used to monitor the chip itself, so it does not give the ambient temperature of the air around the CB. When this register is read, you need to divide the number by 10 to get the temperature in degrees Celsius.

4.6 Current Monitor

There is an INA219 current monitor on the SoNDe CB's 5V rail (after the PoE circuit). It is connected to a 10 mΩ shunt resistor. Through system registers in the *Monitoring* category, the bus voltage, power consumption and current draw can be read.

Two read operations must be done to read out one of the monitor's functions. The conversions from the resulting numbers into relatable units are as follows:

- Bus voltage: Shift the number 3 units to the right and multiply with 0.004 to get the number in volts.
- Power: Multiply the number with 0.02 to get the power consumption in watts.
- Current: No conversion needed. Current is presented in mA.

Note: The current monitor unit is not calibrated. Small deviations from actual values may occur.

4.7 Dummy Data Generator

The dummy data generator produces fabricated data that can be transmitted in either of the three readout modes. This feature generates the data at the FPGA, and hence tests the connection between the FPGA and the SoM. The data is generated at a fixed rate of 60 events/s and continues until the dummy data generator is deactivated. When used, the channels of the front-end should be disabled.

The dummy data generator has two modes:

1. The channel is fixed (channel 100), and the ADC data ramps up 1 LSB per event.
2. The ADC data is fixed (1024 LSB), and the channel ramps up by 1 per event.

When the maximum is reached in either of the two modes, the generator starts over again.

4.8 Synchronization

There are three connectors that can be used for synchronization:

- **Sync In:** SMB connector on bottom side of board.
- **Sync Out:** SMB connector on top side of board.
- **Sync MLVDS:** Micromatch connector on bottom side of board.

The SMB connectors operate with LVCMOS (3.3V) signals, and the Micromatch connector operates with differential signals (common mode 0.9V).

Sync In is connected to the timestamp counter. When a rising edge is detected on the input, the timestamp is immediately reset. If this happens during the gathering of events for a data word, the event gathering is disrupted, and the data is sent, directly followed by a timestamp reset data word. The data word that was disrupted will be completed with zeroes, making it the same size as regular data words.

Sync Out mimics the signal on **Sync In**. The input signal is passed through a buffer and sent to the output connector immediately.

When using **MLVDS**, the DIP switch marked by SW1 on the SoNDe CB must be set accordingly:

- Master board: SW1.1 = ON, SW1.2 = ON
- Slave board: SW1.1 = OFF, SW1.2 = OFF
- Last slave board in a chain: SW1.1 = ON, SW1.2 = OFF

MLVDS has different functionality whether or not the board is a Master board:

- When the board is Master: The MLVDS output mimics incoming pulses on **Sync In**.
- When the board is not Master: The board treats the MLVDS input identically to how **Sync In** pulses are treated, by resetting the timestamp.

4.9 Using the SoNDe CB with IDEAS Testbench

IDEAS Testbench is currently functioning with all-channel readout, while it is very limited for trigger time readout. This is due to the two modes needing two different GUIs, where the trigger time GUI is not adapted to the increased number of channels.

Five scripts are included:

- **external_charges.py** sets up the system for either of the two modes. You change the mode by entering either **triggertime** or **pulseheight** or **single-pulseheight** into the **mode** user setting. After executing the script, acquisition will begin, and all channels are active.
- **gain.py** uses the calibration pulse generator to apply charges to a given number of channels. By entering **256** into the **channels** user setting, all channels are pulsed. You can also specify the two DAC values to use. Before the calibration pulsing, a configurable number of forced readouts (no charge) is applied, to obtain a baseline. *Only works in all-channel mode.*
- **cal_pulsing.py** starts sending calibration charges with a chosen DAC value to a chosen channel. The number of pulses is without limit. *Only works in all-channel mode.*

- **monitoring.py** does a single readout of all monitoring functions.
- **dummy_data.py** disables all the front-end channels and starts sending dummy data in a chosen mode, selected with the parameters in the user settings section.

Important: Scripts can currently not be run while using the trigger time GUI. If you wish to activate trigger time, this must first be done by executing the script in the pulse-height GUI, and then you switch to the other GUI.

4.10 Data Format

Data formats are described in full in the IDEAS packet protocol reference [IDE-REP-Ref-V1.7]. This chapter highlights the data packets that are most relevant for the SoNDe CB.

The data format is different, depending on which type of data is transmitted. A 10-byte packet header always precedes the data packet, as described in the following section.

4.10.1 Packet Header

The Packet Header format is a fixed length number of parameters. The parameters differ slightly depending on the source of the packet.

Table 2: Packet header - System to PC packets.

Packet Header						
Packet ID			Packet Sequence		Timestamp	Data Length
Version	System Number	Packet Type	Sequence Flag	Packet Count		
Fixed	Enum	Enum	Enum	Uint14	Uint32	Uint16
3 bits	5 bits	8 bits	2 bits	14 bits	32 bits	16 bits
2 bytes			2 bytes		4 bytes	2 bytes
10 bytes						

Table 3: Packet header - PC to system packets.

Packet Header						
Packet ID			Packet Sequence		Reserved	Data Length
Version	System Number	Packet Type	Sequence Flag	Packet Count		
Fixed	Enum	Enum	Enum	Uint14	Fixed = 0	Uint16
3 bits	5 bits	8 bits	2 bits	14 bits	32 bits	16 bits
2 bytes			2 bytes		4 bytes	2 bytes
10 bytes						

Packet Header field descriptions:

- Packet ID field
 - Version – Version of packet protocol. 3 bits fixed value = 0b000.
 - System Number – unique ID number of system.

- Packet type – specifies how the “Packet data” field should be decoded (see sections below).
- Packet Sequence field
 - Packet sequence – specifies what sequence (e.g. for read-out data, there will be a series of packets all in the same sequence) the packet is a part of:
 - 0b00 – *StandAlone* – packet is the only packet in the sequence.
 - 0b01 – *FirstPacket* – packet is the first packet in a sequence.
 - 0b10 – *ContinuationPacket* – packet continues a started sequence.
 - 0b11 – *LastPacket* – packet is the last packet in the sequence.
 - Packet count – Incremental counter for packet identification (one counter per system). Values: 0 to 0x3FFF
- Time stamp – System timestamp when packet was created. Values 0-2³².
- Data length – Defines the length of the “Packet Data” field in number of bytes.

4.10.2 Trigger Time

The trigger time mode uses the **D6** data format (same as on S-DAM and ROSMAP-MP). The number of events gathered per packet can be configured up to 250 (higher event numbers give higher throughput. After events for a packet has started being gathered, there is a 5 second timeout, which will result in the remainder of the packet being filled with zeroes. For this application, the 2+6 bits reserved for ASIC+channel are instead used as 8 bits for channel ID (0-255).

Table 4: Trigger time data packet.

Data packet					
Number of events	Event 1 timestamp	Event 1 triggered ASIC + channel	...	Event N timestamp	Event N triggered ASIC + channel
1 byte	4 bytes	1 byte (2+6 bits)		4 bytes	1 byte (2+6 bits)
1 + 5*N bytes					

4.10.3 All-channel Readout

All-channel readout uses the **D4** data format (same format as one-channel readout for S-DAM and ROSMAP-MP). The data is locked to transmitting all 256 channels per data packet. These are labelled with channel number 0-255. Table 5 shows the data format.

Table 5: Multi-event pulse-height data packet.

Data packet										
Packet Data Field Header		Event #1						...	Event #N	
Number of Events (N)	Number of Samples / Event (M)	Event Header	Event Sample Data #1				...			Event Sample Data #M
		Timestamp	Trigger Type	Source ID	Channel ID	Sample				
		1 byte	1 byte	1 byte	2 bytes					
1 byte	2 bytes	4 bytes	5 bytes				5 bytes			
3 bytes		4 + 5*M bytes						4 + 5*M bytes		

	$(4 + 5 \cdot M) \cdot N$ bytes
12 – 1450 bytes	

The following parameters are used in the data packets for the SoNDe CB:

- Number of events – always set to 1
- Number of samples per event – always set to 256
- Source ID – always set to 0
- Channel ID – 0 to 255
- Sample – 12bit ADC data

4.10.4 One-channel Readout

One-channel readout uses the **D4** data format (same format as one-channel readout for S-DAM and ROSMAP-MP). The data is locked to transmitting one channel per data packet. Table 5 in the previous subsection shows the data format.

The following parameters are used in the data packets for the SoNDe CB:

- Number of events – always set to 1
- Number of samples/events – always set to 1
- Source ID – always set to 0
- Channel ID – 0 to 255
- Sample – 12bit ADC data

4.10.5 Timestamp Data Packets

There are two readout packets that mark the wrap-around and reset of the internal timestamp. A packet is sent when the user issues a timestamp reset command, and a packet is also sent when the timestamp counter reaches maximum and starts over again (after approximately 43 seconds).

The timestamp data packets differ based on the readout mode that is active:

- When in trigger time mode, you receive a trigger time data packet with timestamp zero and channel ID for all events in the packet following the code below.
- When in all-channel readout mode, you receive an all-channel readout data packet with timestamp zero and all Channel ID fields being identical, using the code below.
- When in one-channel readout mode, you receive an all-channel readout data packet with timestamp zero and all Channel ID fields being identical, using the code below.

Wrap-around: Data packet using Channel ID **1**

Timestamp reset command: Data packet using Channel ID **2**

If a timestamp reset or wrap-around occurs during the gathering of data for a multi-event data word, the gathering will be disrupted, filling out the rest of the word with zeroes. This data word is then transmitted, immediately followed by a timestamp data packet.

5 Configuration Registers

This section provides a list of the registers that can be set for controlling the SoNDe CB and its front-ends. The registers are organized in two groups: The front-end registers that are used to individually configure the 16 ASICs, and the back-end registers that are used to control the system-type functions like readout and calibration settings.

5.1 Front-end Settings

The front-end settings are listed in Table 6. The default values are here recommended default values. When the front-ends boot up, the configuration register contents are random.

Table 6: Configuration for each ASIC on the front-ends.

Name	Bit No.	No. of Bits	Default	Notes
Creg_lg_prebi_boost	1	1	0	Not in use
Creg_dac_ibi_add_5u0	2	1	0	Not in use
Creg_dac_ibi_add_7u5	3	1	0	Not in use
Pos_thr	4	1	1	Logic high enables the triggering on a negative threshold and inverts the polarity of threshold DACs.
Tout_dis	5	1	0	Not in use
Logic_trigger	6	1	0	Must be 0. Do not touch.
Calint-bypass	7	1	0	Logic high bypasses the internal test capacitor on the CAL input.
HG_0_11	8	1	0	Not in use
HG_12_15	9	1	0	Not in use
Test_on	10	1	0	Logic high enables the CAL input i.e. testing the circuit by injecting test charges is possible via the CAL input (1 pF). Must be logic low if charges are injected directly into the input connectors of the module.
ERO	11	1	0	Should not be used. (Logic high forces the analogue output buffer to be active.)
DLT_en	12	1	0	Should not be used. (Logic high disables late triggers during analogue readout). Default: active.
CHANNEL DISABLE REGISTER [0]	13:14	2	0	Two-bit channel disable register. One bit for each trigger threshold. Second bit (higher bit-number) corresponds to the high range DAC. Only the high range threshold DACs are used. Set value to "2" to disable a channel.
CHANNEL DISABLE REGISTER [1]	15:16	2	0	See above.
CHANNEL DISABLE REGISTER [2]	17:18	2	0	See above.

CHANNEL DISABLE REGISTER [3]	19:20	2	0	See above.
CHANNEL DISABLE REGISTER [4]	21:22	2	0	See above.
CHANNEL DISABLE REGISTER [5]	23:24	2	0	See above.
CHANNEL DISABLE REGISTER [6]	25:26	2	0	See above.
CHANNEL DISABLE REGISTER [7]	27:28	2	0	See above.
CHANNEL DISABLE REGISTER [8]	29:30	2	0	See above.
CHANNEL DISABLE REGISTER [9]	31:32	2	0	See above.
CHANNEL DISABLE REGISTER [10]	33:34	2	0	See above.
CHANNEL DISABLE REGISTER [11]	35:36	2	0	See above.
CHANNEL DISABLE REGISTER [12]	37:38	2	0	See above.
CHANNEL DISABLE REGISTER [13]	39:40	2	0	See above.
CHANNEL DISABLE REGISTER [14]	41:42	2	0	See above.
CHANNEL DISABLE REGISTER [15]	43:44	2	0	See above.
CHANNEL DISABLE REGISTER [16]	45	1	0	Not in use
CHANNEL DISABLE REGISTER [17]	46	1	0	Not in use
CHANNEL DISABLE REGISTER [18]	47	1	0	Not in use
CHANNEL DISABLE REGISTER [19]	48	1	0	Not in use
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [0]	49:56	8	30	Not in use. Set to 30 as default.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [0]	57:64	8	30	8 Bit threshold of the respective channel. Range ca. between 1% and 75% of input DNR.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [1]	65:72	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [1]	73:80	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [2]	81:88	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [2]	89:96	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [3]	97:104	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [3]	105:112	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [4]	113:120	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [4]	121:128	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [5]	129:136	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [5]	137:144	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [6]	145:152	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [6]	153:160	8	30	See above.

HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [7]	161:168	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [7]	169:176	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [8]	177:184	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [8]	185:192	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [9]	193:200	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [9]	201:208	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [10]	209:216	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [10]	217:224	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [11]	225:232	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [11]	233:240	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [12]	241:248	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [12]	249:256	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [13]	257:264	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [13]	265:272	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [14]	273:280	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [14]	281:288	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC LOW RANGE [15]	289:296	8	30	See above.
HIGH GAIN THRESHOLD TRIM DAC HIGH RANGE [15]	297:304	8	30	See above.
LOW GAIN THRESHOLD TRIM DAC [16]	305:312	8	30	Not in use. Set to 30 as default.
LOW GAIN THRESHOLD TRIM DAC [17]	313:320	8	30	Not in use. Set to 30 as default.
LOW GAIN THRESHOLD TRIM DAC [18]	321:328	8	30	Not in use. Set to 30 as default.
LOW GAIN THRESHOLD TRIM DAC [19]	329:336	8	30	Not in use. Set to 30 as default.
TEST ENABLE [0]	337	1	0	Logic high enables charge injection through the CAL input. The test_on bit (bit 10) also must be set logical high. Only one channel should be active at a given time. With more than one channel active, the calibration is not meaningful.
TEST ENABLE [1]	338	1	0	See above.
TEST ENABLE [2]	339	1	0	See above.
TEST ENABLE [3]	340	1	0	See above.

TEST ENABLE [4]	341	1	0	See above.
TEST ENABLE [5]	342	1	0	See above.
TEST ENABLE [6]	343	1	0	See above.
TEST ENABLE [7]	344	1	0	See above.
TEST ENABLE [8]	345	1	0	See above.
TEST ENABLE [9]	346	1	0	See above.
TEST ENABLE [10]	347	1	0	See above.
TEST ENABLE [11]	348	1	0	See above.
TEST ENABLE [12]	349	1	0	See above.
TEST ENABLE [13]	350	1	0	See above.
TEST ENABLE [14]	351	1	0	See above.
TEST ENABLE [15]	352	1	0	See above.
TEST ENABLE [16]	353	1	0	See above.
TEST ENABLE [17]	354	1	0	See above.
TEST ENABLE [18]	355	1	0	See above.
TEST ENABLE [19]	356	1	0	See above.

5.2 Back-end Settings

Table 7 Back-end configuration settings.

Category	Register name	Address	Size in bits	Read/Write	Description
System	SerialNumber	0x0000	32	R	Unique serial number of unit.
System	FirmwareType	0x0001	16	R	Type of firmware
System	FirmwareVersion	0x0002	16	R	Firmware version number.
System	SystemNumber	0x0010	5	R/W	System ID
System	MAC address	0x0030	48	R	MAC address of the CB
Board	Front-end power enable	0x0B00	1	R/W	Enable/disable front-end power rails
Board	Ethernet LED setting	0x0B01	1	R/W	0: standard (based on Ethernet activity) 1: debug (based on LED debug reg.)
Board	Ethernet LED debug	0x0B02	1	R/W	0: both off 1: both on
Cal Pulse Gen	Calibration Execute	0x0C00	1	P	Start calibration pulsing with current settings.
Cal Pulse Gen	Calibration Num Pulses	0x0C02	32	R/W	Number of pulses to generate.
Cal Pulse Gen	Calibration Pulse Length	0x0C03	32	R/W	Length of cal pulses in number of system clock cycles. Actual length is +1
Cal Pulse Gen	Calibration Pulse Interval	0x0C04	32	R/W	Clock cycles between start of each cal pulse. Actual length is +1.
Cal Pulse Gen	Calibration Pulse DAC	0x0E00	12	R/W	DAC for setting the calibration pulse amplitude.
Detector Bias	HV DAC 1	0x0E01	12	R/W	DAC for setting the HV bias.

Detector Bias	HV DAC 2	0x0E02	12	R/W	DAC for setting the HV bias.
Detector Bias	HV DAC 3	0x0E03	12	R/W	DAC for setting the HV bias.
Detector Bias	HV DAC 4	0x0E04	12	R/W	DAC for setting the HV bias.
Monitoring	SoC Temperature	0x0E10	12	R	Current temperature on SoC chip
Monitoring	FE 1 temperature	0x0F00	12	R	Temperature for front-end 1
Monitoring	FE 2 temperature	0x0F01	12	R	Temperature for front-end 2
Monitoring	FE 3 temperature	0x0F02	12	R	Temperature for front-end 3
Monitoring	FE 4 temperature	0x0F03	12	R	Temperature for front-end 4
Monitoring	Bus voltage	0x0F06	12	R	5V bus voltage measurement
Monitoring	Power consumption	0x0F07	12	R	System power consumption
Monitoring	Current	0x0F08	12	R	System current draw
Readout	Readout enable	0xF009	3	R/W	011: trigger time 101: pulseheight 111: single-pulseheight
Readout	Readout reset	0xF00A	1	R/W	Resets readout
Readout	Hold delay	0xF00B	16	R/W	Sets hold delay. Different for cal and ext. pulsing. 1.45 us: 292 (cal), 264 (ext). 5 ns per LSB
Readout	Reset timestamp	0xF00C	1	R/W	Resets timestamp
Readout	Dummy data generator	0xF00D	2	R/W	00: off 01: fixed ch, increased ADC 10: ramped ch, fixed ADC
Readout	Trigger time number of events	0xF00E	16	R/W	Number of events collated in trigger time packet

5.3 Hamamatsu H12700 Documentation

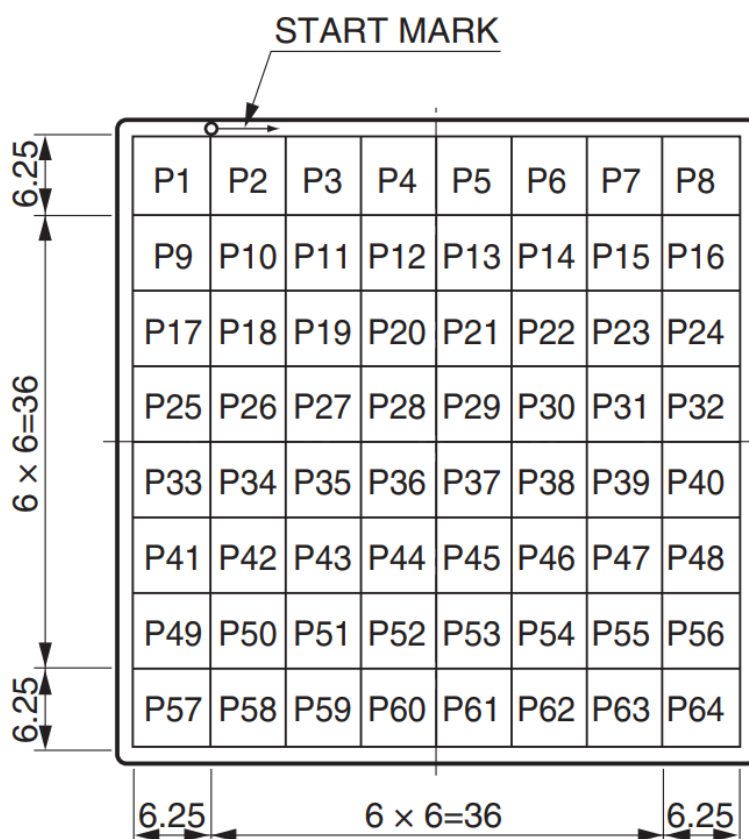


Figure 4: Hamamatsu H12700 pixel map.

CONNECTION FOR SIGNAL CONNECTORS (BOTTOM VIEW)

GND	P7	GND	P5	GND	P3	GND	P1
GND	P8	GND	P6	GND	P4	GND	P2
GND	P15	GND	P13	GND	P11	GND	P9
GND	P16	GND	P14	GND	P12	GND	P10
GND	P23	GND	P21	GND	P19	GND	P17
GND	P24	GND	P22	GND	P20	GND	P18
GND	P31	GND	P29	GND	P27	GND	P25
GND	P32	GND	P30	GND	P28	GND	P26
GND	P39	GND	P37	GND	P35	GND	P33
GND	P40	GND	P38	GND	P36	GND	P34
GND	P47	GND	P45	GND	P43	GND	P41
GND	P48	GND	P46	GND	P44	GND	P42
GND	P55	GND	P53	GND	P51	GND	P49
GND	P56	GND	P54	GND	P52	GND	P50
GND	P63	GND	P61	GND	P59	GND	P57
GND	P64	GND	P62	GND	P60	GND	P58
GND	DY10	GND	GND	*B	GND	GND	*B
*B	GND	GND	*B	GND	GND	GND	GND
SIG4		SIG3		SIG2		SIG1	

NOTE *B: No pin.

Figure 5: Hamamatsu H12700 connector pinout.

5.4 SoNDe CB Pixel Map

The pixel map containing all four front-ends of a SoNDe CB is shown in Figure 6. The orientation/viewpoint is the same as in Figure 4. The rows and column that correspond to each pixel for each of the front-end are also shown.

ROW	15	P1	P2	P3	P4	P5	P6	P7	P8	P1	P2	P3	P4	P5	P6	P7	P8				
	14	P9	P10	P11	P12	P13	P14	P15	P16	P9	P10	P11	P12	P13	P14	P15	P16				
	13	P17	P18	P19	P20	P21	P22	P23	P24	P17	P18	P19	P20	P21	P22	P23	P24				
	12	P25	P26	P27	P28	P29	P30	P31	P32	P25	P26	P27	P28	P29	P30	P31	P32				
	11	P33	P34	P35	P36	P37	P38	P39	P40	P33	P34	P35	P36	P37	P38	P39	P40				
	10	P41	P42	P43	P44	P45	P46	P47	P48	P41	P42	P43	P44	P45	P46	P47	P48				
	9	P49	P50	P51	P52	P53	P54	P55	P56	P49	P50	P51	P52	P53	P54	P55	P56				
	8	P57	P58	P59	P60	P61	P62	P63	P64	P57	P58	P59	P60	P61	P62	P63	P64				
	7	P1	P2	P3	P4	P5	P6	P7	P8	P1	P2	P3	P4	P5	P6	P7	P8				
	6	P9	P10	P11	P12	P13	P14	P15	P16	P9	P10	P11	P12	P13	P14	P15	P16				
	5	P17	P18	P19	P20	P21	P22	P23	P24	P17	P18	P19	P20	P21	P22	P23	P24				
	4	P25	P26	P27	P28	P29	P30	P31	P32	P25	P26	P27	P28	P29	P30	P31	P32				
	3	P33	P34	P35	P36	P37	P38	P39	P40	P33	P34	P35	P36	P37	P38	P39	P40				
	2	P41	P42	P43	P44	P45	P46	P47	P48	P41	P42	P43	P44	P45	P46	P47	P48				
	1	P49	P50	P51	P52	P53	P54	P55	P56	P49	P50	P51	P52	P53	P54	P55	P56				
	0	P57	P58	P59	P60	P61	P62	P63	P64	P57	P58	P59	P60	P61	P62	P63	P64				
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15					
COLUMN																					
									<table><tr><td>FE2</td><td>FE0</td></tr><tr><td>FE3</td><td>FE1</td></tr></table>		FE2	FE0	FE3	FE1							
FE2	FE0																				
FE3	FE1																				

Figure 6: SoNDe CB pixel map.

5.5 Channel to Pixel Conversion

Table 8 lists all the channels of a SoNDe CB and which pixel they correspond to:

- **Data channel (0-255):** The channel numbers that appear in the data stream.
- **Front-end (0-3):** Front-ends that connect to the MaPMTs.
- **PMT channel (1-64):** Pixel ID as defined in Figure 4.
- **ASIC (0-3):** ASIC ID for a given front-end.
- **ASIC channel (0-15):** Channel ID for a given ASIC.
- **Column (0-15):** Column in pixel map encompassing four front-ends.
- **Row (0-15):** Row in pixel map encompassing four front-ends.

Table 8: Channel to pixel conversion.

Data channel	Front-end	PMT channel	ASIC	ASIC channel	Column	Row
0	0	32	0	0	15	12
1	0	31	0	1	14	12
2	0	30	0	2	13	12
3	0	29	0	3	12	12
4	0	24	0	4	15	13
5	0	23	0	5	14	13
6	0	22	0	6	13	13
7	0	21	0	7	12	13
8	0	16	0	8	15	14
9	0	15	0	9	14	14
10	0	14	0	10	13	14
11	0	13	0	11	12	14
12	0	8	0	12	15	15
13	0	7	0	13	14	15
14	0	6	0	14	13	15
15	0	5	0	15	12	15
16	0	1	1	0	8	15
17	0	2	1	1	9	15
18	0	3	1	2	10	15
19	0	4	1	3	11	15
20	0	9	1	4	8	14
21	0	10	1	5	9	14
22	0	11	1	6	10	14
23	0	12	1	7	11	14
24	0	17	1	8	8	13
25	0	18	1	9	9	13
26	0	19	1	10	10	13
27	0	20	1	11	11	13
28	0	25	1	12	8	12
29	0	26	1	13	9	12
30	0	27	1	14	10	12
31	0	28	1	15	11	12

32	0	64	2	0	15	8
33	0	63	2	1	14	8
34	0	62	2	2	13	8
35	0	61	2	3	12	8
36	0	56	2	4	15	9
37	0	55	2	5	14	9
38	0	54	2	6	13	9
39	0	53	2	7	12	9
40	0	48	2	8	15	10
41	0	47	2	9	14	10
42	0	46	2	10	13	10
43	0	45	2	11	12	10
44	0	40	2	12	15	11
45	0	39	2	13	14	11
46	0	38	2	14	13	11
47	0	37	2	15	12	11
48	0	33	3	0	8	11
49	0	34	3	1	9	11
50	0	35	3	2	10	11
51	0	36	3	3	11	11
52	0	41	3	4	8	10
53	0	42	3	5	9	10
54	0	43	3	6	10	10
55	0	44	3	7	11	10
56	0	49	3	8	8	9
57	0	50	3	9	9	9
58	0	51	3	10	10	9
59	0	52	3	11	11	9
60	0	57	3	12	8	8
61	0	58	3	13	9	8
62	0	59	3	14	10	8
63	0	60	3	15	11	8
64	1	32	0	0	15	4
65	1	31	0	1	14	4
66	1	30	0	2	13	4
67	1	29	0	3	12	4
68	1	24	0	4	15	5
69	1	23	0	5	14	5
70	1	22	0	6	13	5
71	1	21	0	7	12	5
72	1	16	0	8	15	6
73	1	15	0	9	14	6
74	1	14	0	10	13	6
75	1	13	0	11	12	6
76	1	8	0	12	15	7
77	1	7	0	13	14	7
78	1	6	0	14	13	7
79	1	5	0	15	12	7

80	1	1	1	0	8	7
81	1	2	1	1	9	7
82	1	3	1	2	10	7
83	1	4	1	3	11	7
84	1	9	1	4	8	6
85	1	10	1	5	9	6
86	1	11	1	6	10	6
87	1	12	1	7	11	6
88	1	17	1	8	8	5
89	1	18	1	9	9	5
90	1	19	1	10	10	5
91	1	20	1	11	11	5
92	1	25	1	12	8	4
93	1	26	1	13	9	4
94	1	27	1	14	10	4
95	1	28	1	15	11	4
96	1	64	2	0	15	0
97	1	63	2	1	14	0
98	1	62	2	2	13	0
99	1	61	2	3	12	0
100	1	56	2	4	15	1
101	1	55	2	5	14	1
102	1	54	2	6	13	1
103	1	53	2	7	12	1
104	1	48	2	8	15	2
105	1	47	2	9	14	2
106	1	46	2	10	13	2
107	1	45	2	11	12	2
108	1	40	2	12	15	3
109	1	39	2	13	14	3
110	1	38	2	14	13	3
111	1	37	2	15	12	3
112	1	33	3	0	8	3
113	1	34	3	1	9	3
114	1	35	3	2	10	3
115	1	36	3	3	11	3
116	1	41	3	4	8	2
117	1	42	3	5	9	2
118	1	43	3	6	10	2
119	1	44	3	7	11	2
120	1	49	3	8	8	1
121	1	50	3	9	9	1
122	1	51	3	10	10	1
123	1	52	3	11	11	1
124	1	57	3	12	8	0
125	1	58	3	13	9	0
126	1	59	3	14	10	0
127	1	60	3	15	11	0

128	2	32	0	0	7	12
129	2	31	0	1	6	12
130	2	30	0	2	5	12
131	2	29	0	3	4	12
132	2	24	0	4	7	13
133	2	23	0	5	6	13
134	2	22	0	6	5	13
135	2	21	0	7	4	13
136	2	16	0	8	7	14
137	2	15	0	9	6	14
138	2	14	0	10	5	14
139	2	13	0	11	4	14
140	2	8	0	12	7	15
141	2	7	0	13	6	15
142	2	6	0	14	5	15
143	2	5	0	15	4	15
144	2	1	1	0	0	15
145	2	2	1	1	1	15
146	2	3	1	2	2	15
147	2	4	1	3	3	15
148	2	9	1	4	0	14
149	2	10	1	5	1	14
150	2	11	1	6	2	14
151	2	12	1	7	3	14
152	2	17	1	8	0	13
153	2	18	1	9	1	13
154	2	19	1	10	2	13
155	2	20	1	11	3	13
156	2	25	1	12	0	12
157	2	26	1	13	1	12
158	2	27	1	14	2	12
159	2	28	1	15	3	12
160	2	64	2	0	7	8
161	2	63	2	1	6	8
162	2	62	2	2	5	8
163	2	61	2	3	4	8
164	2	56	2	4	7	9
165	2	55	2	5	6	9
166	2	54	2	6	5	9
167	2	53	2	7	4	9
168	2	48	2	8	7	10
169	2	47	2	9	6	10
170	2	46	2	10	5	10
171	2	45	2	11	4	10
172	2	40	2	12	7	11
173	2	39	2	13	6	11
174	2	38	2	14	5	11
175	2	37	2	15	4	11

176	2	33	3	0	0	11
177	2	34	3	1	1	11
178	2	35	3	2	2	11
179	2	36	3	3	3	11
180	2	41	3	4	0	10
181	2	42	3	5	1	10
182	2	43	3	6	2	10
183	2	44	3	7	3	10
184	2	49	3	8	0	9
185	2	50	3	9	1	9
186	2	51	3	10	2	9
187	2	52	3	11	3	9
188	2	57	3	12	0	8
189	2	58	3	13	1	8
190	2	59	3	14	2	8
191	2	60	3	15	3	8
192	3	32	0	0	7	4
193	3	31	0	1	6	4
194	3	30	0	2	5	4
195	3	29	0	3	4	4
196	3	24	0	4	7	5
197	3	23	0	5	6	5
198	3	22	0	6	5	5
199	3	21	0	7	4	5
200	3	16	0	8	7	6
201	3	15	0	9	6	6
202	3	14	0	10	5	6
203	3	13	0	11	4	6
204	3	8	0	12	7	7
205	3	7	0	13	6	7
206	3	6	0	14	5	7
207	3	5	0	15	4	7
208	3	1	1	0	0	7
209	3	2	1	1	1	7
210	3	3	1	2	2	7
211	3	4	1	3	3	7
212	3	9	1	4	0	6
213	3	10	1	5	1	6
214	3	11	1	6	2	6
215	3	12	1	7	3	6
216	3	17	1	8	0	5
217	3	18	1	9	1	5
218	3	19	1	10	2	5
219	3	20	1	11	3	5
220	3	25	1	12	0	4
221	3	26	1	13	1	4
222	3	27	1	14	2	4
223	3	28	1	15	3	4

224	3	64	2	0	7	0
225	3	63	2	1	6	0
226	3	62	2	2	5	0
227	3	61	2	3	4	0
228	3	56	2	4	7	1
229	3	55	2	5	6	1
230	3	54	2	6	5	1
231	3	53	2	7	4	1
232	3	48	2	8	7	2
233	3	47	2	9	6	2
234	3	46	2	10	5	2
235	3	45	2	11	4	2
236	3	40	2	12	7	3
237	3	39	2	13	6	3
238	3	38	2	14	5	3
239	3	37	2	15	4	3
240	3	33	3	0	0	3
241	3	34	3	1	1	3
242	3	35	3	2	2	3
243	3	36	3	3	3	3
244	3	41	3	4	0	2
245	3	42	3	5	1	2
246	3	43	3	6	2	2
247	3	44	3	7	3	2
248	3	49	3	8	0	1
249	3	50	3	9	1	1
250	3	51	3	10	2	1
251	3	52	3	11	3	1
252	3	57	3	12	0	0
253	3	58	3	13	1	0
254	3	59	3	14	2	0
255	3	60	3	15	3	0

6 Known Bugs

- When getting data above the maximum data rates, the timestamp wrap packets may be lost.
- When attempting to read from a temperature sensor that is not connected, connection is lost.